Lab 2: ASP.NET Core Web API with Swagger

Software Required

- Visual Studio 2022 Community (Purple Icon)
- .NET 8 SDK
- Postman (Later)

Step 1: Open Visual Studio

1. Open **Visual Studio 2022 Community**.
2. Click **Create a new project**.

Step 2: Select Project Template

In the search box, type:

ASP.NET Core Web API

Select:

ASP.NET Core Web API

Click **Next**.

Step 3: Configure Your Project

Project Name:

SwaggerDemo

Location:

Choose any folder.

Solution Name:

SwaggerDemo

Click **Next**.

Step 4: Additional Information

Select the following:

Option	Value
Framework	**.NET 8.0 (Long Term Support)**
Authentication Type	**None**
Configure for HTTPS	✓
Enable OpenAPI Support	✓
Use Controllers	✓
Native AOT	✗

Click **Create**.

Step 5: Project Structure

Visual Studio creates the following files:

```
SwaggerDemo
|
 ├── Controllers
|     └── WeatherForecastController.cs
|
 ├── Properties
|     └── launchSettings.json
|
 ├── appsettings.json
|
 ├── Program.cs
|
 ├── WeatherForecast.cs
|
└── SwaggerDemo.csproj
```

Step 6: Run the Project

Press:

Ctrl + F5

The first time, you may see:

Trust ASP.NET Core SSL Certificate?

Click:

Yes

A browser opens:

https://localhost:xxxx/swagger

You should see the Swagger page.

If Swagger opens, your project is correct.

Step 7: Delete Default Files

Delete:

Controllers
    WeatherForecastController.cs

Delete:

WeatherForecast.cs

Step 8: Create Employee Controller

Right-click:

Controllers

↓

Add

↓

Class

Name:

EmployeeController.cs

Click **Add**.

Step 9: Write Employee Controller

Replace everything with this code.

```
using Microsoft.AspNetCore.Mvc;

namespace SwaggerDemo.Controllers
{
   [ApiController]
   [Route("api/[controller]")]
   public class EmployeeController : ControllerBase
   {
      static List<string> employees = new List<string>()
      {
         "John",
         "David",
         "Smith"
      };

      [HttpGet]
      public IActionResult Get()
      {
         return Ok(employees);
      }

      [HttpPost]
      public IActionResult Post(string name)
      {
         employees.Add(name);
         return Ok("Employee Added Successfully");
      }

      [HttpPut]
      public IActionResult Put(int id, string name)
```

```
        {
            employees[id] = name;
            return Ok("Employee Updated Successfully");
        }

        [HttpDelete]
        public IActionResult Delete(int id)
        {
            employees.RemoveAt(id);
            return Ok("Employee Deleted Successfully");
        }
    }
}
```

Save the file.

Step 10: Program.cs

Keep **Program.cs** exactly like this.

```
var builder = WebApplication.CreateBuilder(args);

// Add services to the container.
builder.Services.AddControllers();

builder.Services.AddEndpointsApiExplorer();

builder.Services.AddSwaggerGen();

var app = builder.Build();

// Configure the HTTP request pipeline.
if (app.Environment.IsDevelopment())
{
    app.UseSwagger();

    app.UseSwaggerUI();
}
```

app.UseHttpsRedirection();

app.UseAuthorization();

app.MapControllers();

app.Run();

Do not add any other code yet.

Step 11: Build the Project

Click:

Build

↓

Build Solution

Shortcut:

Ctrl + Shift + B

You should see:

Build succeeded.

Step 12: Run the Project

Press:

Ctrl + F5

Browser opens:

https://localhost:xxxx/swagger

You should see:

Employee

GET

POST
PUT
DELETE

Step 13: Test GET

Click:

GET

↓

Try it out

↓

Execute

Output:

```
[
  "John",
  "David",
  "Smith"
]
```

Status:

200 OK

Step 14: Test POST

Click:

POST

↓

Try it out

Enter:

James

↓

Execute

Output:

Employee Added Successfully

Step 15: Test PUT

Input:

id = 1

name = Robert

Output:

Employee Updated Successfully

Step 16: Test DELETE

Input:

id = 2

Output:

Employee Deleted Successfully

Step 17: Install Postman

1. Download and install Postman.
2. Open Postman.
3. Click **New → HTTP Request**.
4. Choose **GET**.
5. Enter the URL:

https://localhost:xxxx/api/employee

6. Click **Send**.

You should receive:

```
[
  "John",
  "David",
  "Smith"
]
```

Status:

200 OK

Step 18: Change Route

Open EmployeeController.cs.

Change:

[Route("api/[controller]")]

to:

[Route("api/Emp")]

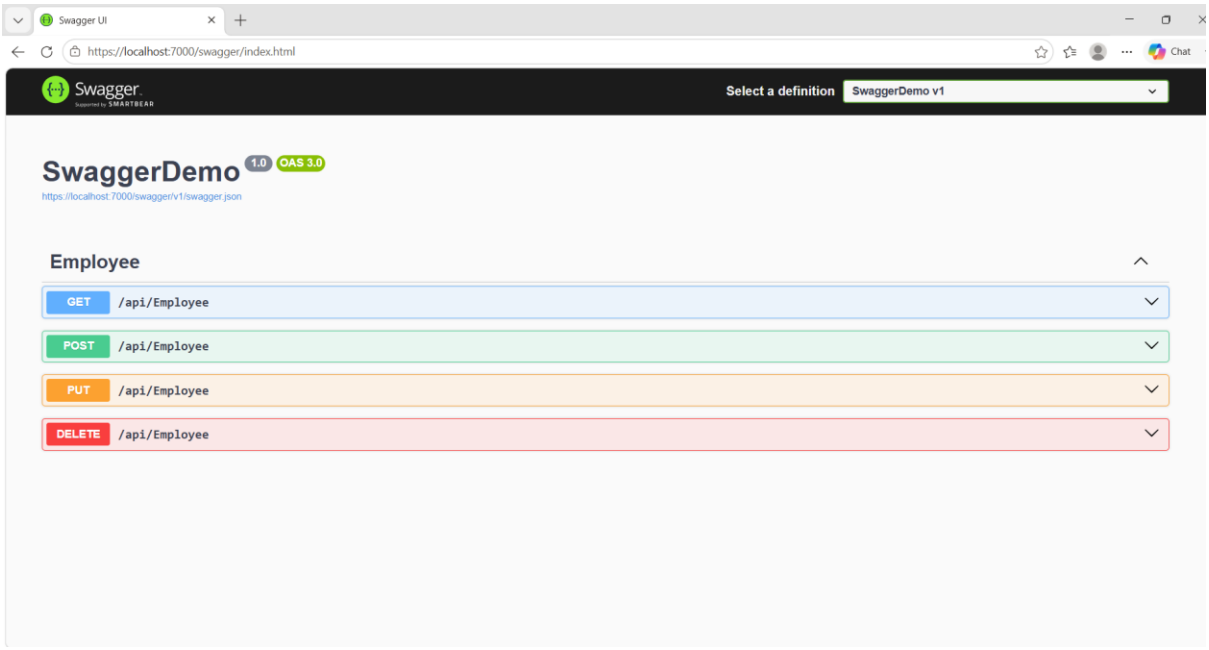
Save the file.

Run the project again.

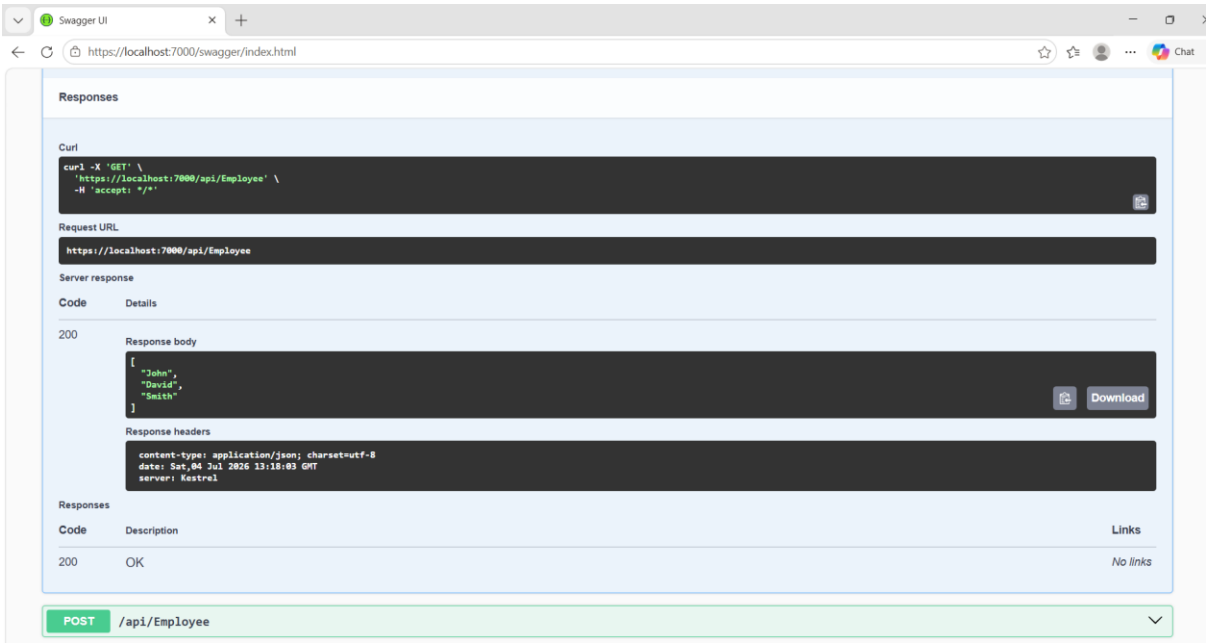
Now use this URL in Swagger or Postman:

https://localhost:xxxx/api/Emp

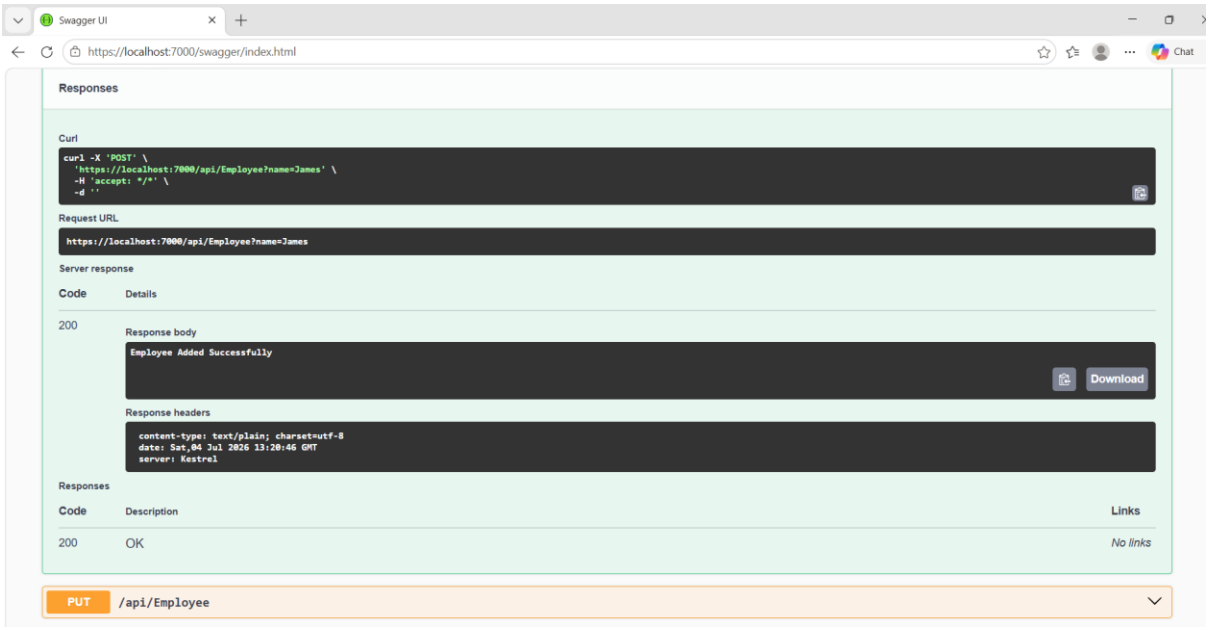
Outputs:



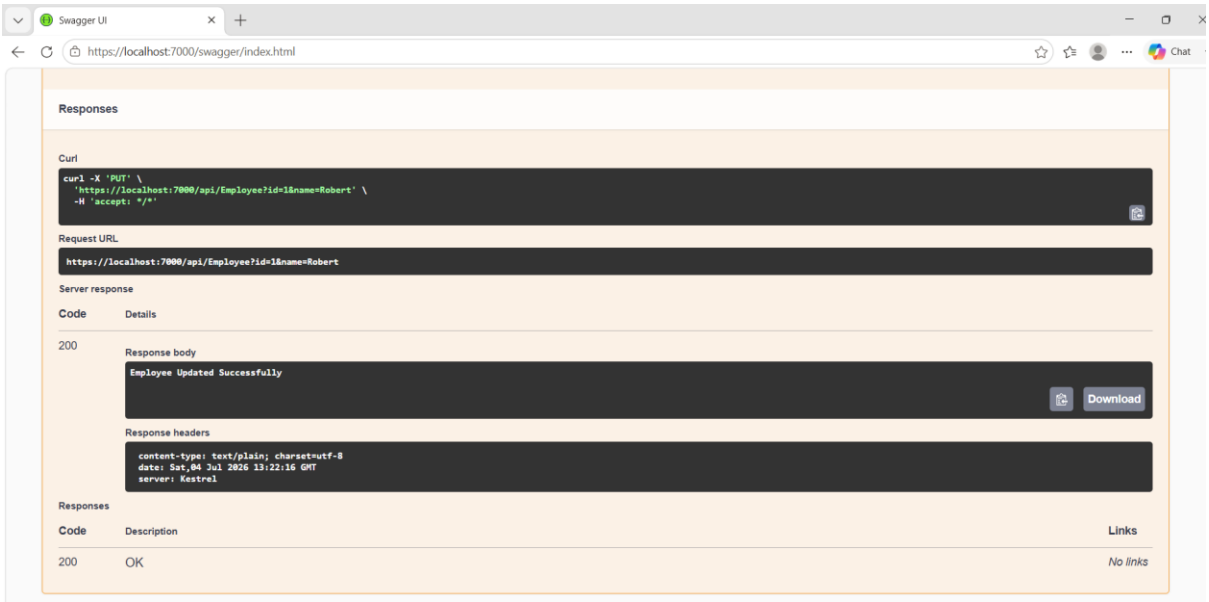
GET METHOD:



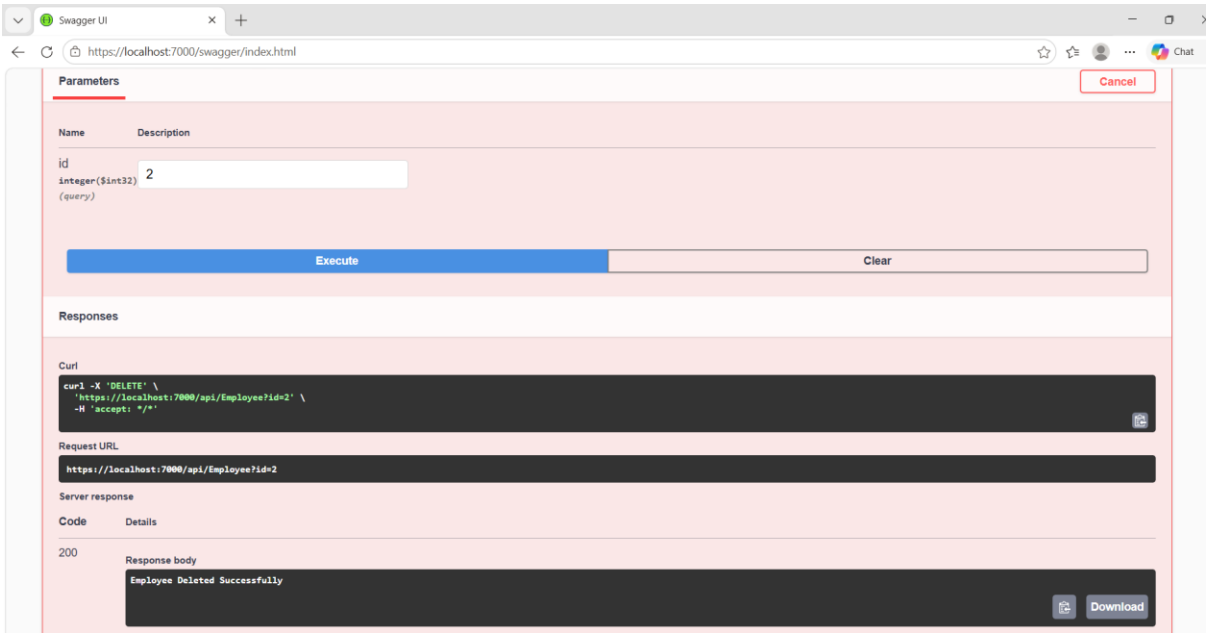
POST METHOD:



POST METHOD:



DELETE METHOD:


Swagger UI
https://localhost:7000/swagger/index.html
Chat
Parameters
Cancel
Name
Description
id
integer($int32)
(query)
2
Execute
Clear
Responses
Curl
curl -X 'DELETE' \
'https://localhost:7000/api/Employee?id=2' \
-H 'accept: */*'
Request URL
https://localhost:7000/api/Employee?id=2
Server response
Code
Details
200
Response body
Employee Deleted Successfully
Download